SAT Math Trick Pack 3

68 new challenge items

Name ..... Date .....

Each question states its tool: **No calculator needed**, **Calculator** or **Desmos**. Multiple-choice questions have one correct answer. For student-produced responses (SPR), write your answer in the box.

Algebra (6 items)

1. A-337 · Hard · Multiple choice · **No calculator needed**

The solution to the system $ax + by = 31$ and $bx + ay = 19$, where a and b are constants, is $(4, 1)$. What is the value of $a^2 - b^2$?

A) 16

B) 40

C) 10

D) 58

2. A-338 · Hard · Multiple choice · **No calculator needed**

In the xy-plane, the graph of $3x - ky = 12$, where k is a nonzero constant, has an x-intercept equal to its y-intercept. What is the value of k?

A) -4

B) 3

C) 4

D) -3

3. A-339 · Hard · Multiple choice · **No calculator needed**

For every real number m, the graph of $y = mx + 3m - 2$ in the xy-plane passes through the point P. What are the coordinates of P?

A) $(-3, -2)$

B) $(0, -2)$

C) $(3, -2)$

D) $(-2, 3)$

4. A-340 · Hard · Multiple choice · **No calculator needed**

A cafe sells juice for 4 dollars each and sandwiches for 7 dollars each. A customer spends exactly 112 dollars and buys at least one juice and at least one sandwich. How many different combinations of juices and sandwiches are possible?

A) 5

B) 4

C) 3

D) 28

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5. **A-341** · Medium · Multiple choice · **No calculator needed**

Which inequality has the solution set $x \leq -3$ or $x \geq 11$?

A) $|x + 4| \geq 7$

B) $|x - 4| \leq 7$

C) $|x - 4| \geq 7$

D) $|x - 7| \geq 4$

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6. **A-342** · Hard · Multiple choice · **No calculator needed**

For how many integer values of k does the point (k, k) satisfy both $y \geq 2x - 5$ and $y > -x + 1$?

A) 5

B) 4

C) Infinitely many

D) 6

Advanced Math (10 items)

7. **AM-322** · Hard · Multiple choice · **No calculator needed**

In the equation $\frac{ax}{x - 2} = 3 + \frac{4}{x - 2}$, a is a constant. For which values of a does the equation have no solution?

A) 3 and 4 only

B) 3 only

C) 2 only

D) 2 and 3 only

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8. **AM-323** · Hard · Multiple choice · **No calculator needed**

The function f is defined by $f(x) = 3x^2 - 5x + 2$. Which expression is equivalent to $f(a + 1) - f(a)$?

A) 0

B) $6a - 2$

C) $6a + 3$

D) $6a + 8$

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9. **AM-324** · Hard · SPR · **No calculator needed**

For how many integer values of b does the equation $x^2 + bx + 36 = 0$ have two distinct integer solutions?

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10. **AM-325** · Hard · Multiple choice · **No calculator needed**

The equations $x^2 - x - 12 = 0$ and $x^2 + kx + 8 = 0$ have a common solution, where k is a positive constant. What is the value of k?

A) -6

B) $\frac{17}{3}$

C) $-\frac{17}{3}$

D) 6

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11. **AM-326** · Hard · Multiple choice · **Desmos**

What is the sum of all distinct real solutions of $x^3 - 2x^2 - 15x + 36 = 0$?

A) 2

B) 7

C) -2

D) -1

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12. **AM-327** · Hard · Multiple choice · **No calculator needed**

What is the solution set of the equation $2x^{2/3} - 3 = 15$?

A) $\{-27, 27\}$

B) $\{27\}$

C) $\{-9^{2/3}, 9^{2/3}\}$

D) $\{-6, 6\}$

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13. **AM-328** · Hard · Multiple choice · **No calculator needed**

The functions f and g are defined by $f(x) = 7 \cdot 4^x$ and $g(x) = f(x - 1.5)$. For all values of x, $g(x) = k \cdot f(x)$, where k is a constant. What is the value of k?

A) -6

B) $\frac{7}{8}$

C) $\frac{1}{8}$

D) 8

14. **AM-329** · Hard · Multiple choice · **No calculator needed**

The function f is defined by $f(x) = 2(x-1)(x-7)$. For what value of k does the graph of $y = f(x)+k$ have exactly one x-intercept?

A) 9

B) 14

C) 18

D) -18

15. **AM-330** · Hard · Multiple choice · **No calculator needed**

The function g is defined by $g(x) = 3^x$. If $g(a) = 4$, what is the value of $g(2a-1)$?

A) $\frac{16}{3}$

B) 15

C) $\frac{8}{3}$

D) 7

16. **AM-331** · Medium · Multiple choice · **No calculator needed**

If $2a - 3b = 7$, what is the value of $\frac{4^a}{8^b}$?

A) $\frac{1}{128}$

B) 2^{14}

C) 7

D) 128

Geometry and Trigonometry (17 items)

17. **GT-258** · Hard · Multiple choice · **No calculator needed**

Right triangles PQR and XYZ are similar, with P, Q and R corresponding to X, Y and Z, and right angles at Q and Y. If $\tan P = \frac{8}{15}$ and $XZ = 51$, what is the length of $\overline{XY}$?

A) 27.2

B) 45

C) 24

D) 95.625

18. GT-259 · Hard · Multiple choice · **No calculator needed**

In triangle ABC, $AB = AC = 17$ and $BC = 16$. What is the value of $\tan B$?

A) $\frac{8}{15}$

B) $\frac{15}{8}$

C) $\frac{15}{17}$

D) $\frac{16}{17}$

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19. GT-260 · Hard · Multiple choice · **No calculator needed**

A rectangular box has a base that measures 9 by 12 and a height of 20. Let θ be the angle between a diagonal of the box (from a bottom corner to the opposite top corner) and the base. What is the value of $\tan\theta$?

A) $\frac{20}{9}$

B) $\frac{5}{3}$

C) $\frac{3}{4}$

D) $\frac{4}{3}$

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20. GT-261 · Hard · Multiple choice · **No calculator needed**

In right triangle ABC, angle C is the right angle and $\cos A = k$. Which expression is equal to $\sin^2 A + \sin^2 B$?

A) 1

B) The value cannot be determined from the information given.

C) $2 - 2k^2$

D) $2k^2$

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21. GT-262 · Hard · Multiple choice · **No calculator needed**

Right triangle ABC has its right angle at C, $AC = 9$ and $BC = 12$. Each leg is lengthened by 3 and the angle at C stays a right angle. What is $\tan A$ in the new triangle?

A) $\frac{4}{5}$

B) $\frac{4}{3}$

C) $\frac{5}{4}$

D) $\frac{5}{3}$

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22. GT-263 · Hard · Multiple choice · **No calculator needed**

A straight railway track up a hill has a constant grade of 75%, meaning it rises 75 meters for every 100 meters of horizontal distance. The track is 300 meters long, measured along the slope. What is the vertical rise of the track, in meters?

A) 400

B) 240

C) 180

D) 225

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23. GT-264 · Hard · Multiple choice · **No calculator needed**

A drone flies 12 kilometers in a direction 60° north of east, then flies 12 kilometers due west. How far, in kilometers, is the drone from its starting point?

A) 12

B) 24

C) $12\sqrt{2}$

D) $6\sqrt{3}$

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24. GT-265 · Hard · Multiple choice · **No calculator needed**

In triangle ABC, the measure of angle B is 45° and the measure of angle C is 30°. The altitude from A to $\overline{BC}$ has length 6. What is the length of $\overline{BC}$?

A) 12

B) $6 + 2\sqrt{3}$

C) $12 + 6\sqrt{2}$

D) $6 + 6\sqrt{3}$

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25. GT-266 · Hard · Multiple choice · **No calculator needed**

In the xy-plane, the points $A(0, 0)$, $B(6, 3)$ and $C(4, 7)$ are the vertices of a triangle. What is the value of $\tan(\angle CAB)$?

A) $\dfrac{1}{2}$

B) $\dfrac{2}{3}$

C) $\dfrac{3}{2}$

D) $\dfrac{7}{4}$

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26. GT-267 · Medium · Multiple choice · **No calculator needed**

A parallelogram has sides of lengths 12 and 20. The cosine of one of its acute angles is 0.8. What is the area of the parallelogram?

A) 72

B) 144

C) 192

D) 240

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27. GT-268 · Hard · Multiple choice · **No calculator needed**

From the top of a cliff 45 meters high, the angles of depression to two boats on opposite sides of the cliff are $30°$ and $45°$. The boats and the base of the cliff are on a straight line. What is the distance, in meters, between the boats?

A) $45\sqrt{3} - 45$

B) $45 + 15\sqrt{3}$

C) $45\sqrt{3}$

D) $45 + 45\sqrt{3}$

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28. GT-269 · Hard · Multiple choice · **No calculator needed**

In a right triangle, the measure of acute angle A is between $45°$ and $90°$. Which of the following is true?

A) $\cos A < \sin A < \tan A$

B) $\tan A < \sin A < \cos A$

C) $\sin A < \cos A < \tan A$

D) $\cos A < \tan A < \sin A$

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29. GT-270 · Hard · SPR · **No calculator needed**

A rectangle has side lengths 6 and $6\sqrt{3}$. What is the measure, in degrees, of the acute angle formed by its diagonals?

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30. GT-271 · Hard · Multiple choice · **No calculator needed**

In the xy-plane, line ℓ has slope $\frac{3}{4}$ and line k is perpendicular to line ℓ. Line k makes an angle θ with the positive x-axis, measured counterclockwise, where $0° < \theta < 180°$. What is the value of $\cos\theta$?

A) $-\frac{4}{5}$

B) $\frac{3}{5}$

C) $-\frac{3}{5}$

D) $\frac{4}{5}$

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31. GT-272 · Hard · Multiple choice · **No calculator needed**

Parallel lines ℓ and m are horizontal, with ℓ above m. Point P lies between the lines. Point A on ℓ and point B on m are both to the right of P. Segment $\overline{PA}$ makes a $38°$ angle with line ℓ, and segment $\overline{PB}$ makes a $47°$ angle with line m, both angles opening toward P. What is the measure, in degrees, of angle APB?

A) 42.5

B) 95

C) 85

D) 9

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32. GT-273 · Hard · Multiple choice · **No calculator needed**

In triangle ABC, the measure of angle A is $52°$. The bisectors of angles B and C intersect at point I. What is the measure, in degrees, of angle BIC?

A) 116

B) 128

C) 64

D) 104

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33. GT-274 · Medium · Multiple choice · **No calculator needed**

In triangle ABC, $AB = 2x + 1$, $BC = x + 6$ and $CA = 3x - 4$. The perimeter of the triangle is 45. Which angle of the triangle has the greatest measure?

A) Angle A

B) It cannot be determined from the information given.

C) Angle C

D) Angle B

Problem-Solving and Data Analysis (35 items)

34. **PS-176** · Hard · Multiple choice · **No calculator needed**

If 18% of a is 24, what is $a\%$ of 18?

A) 4.32

B) 24

C) $2{,}400$

D) $133\frac{1}{3}$

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35. **PS-177** · Medium · Multiple choice · **Calculator**

The monthly rent of an apartment increased by 60%. By what percent must the new rent decrease to return to the original rent?

A) 62.5%

B) 37.5%

C) 60%

D) 40%

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36. **PS-178** · Hard · Multiple choice · **Calculator**

A store's profit on each jacket is 37.5% of the selling price. The profit is what percent of the cost of the jacket to the store?

A) 37.5%

B) 27.3%

C) 62.5%

D) 60%

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37. **PS-179** · Hard · Multiple choice · **Calculator**

A restaurant adds a 15% tax and a 10% service charge to a bill. Both are calculated on the original subtotal. The total paid is 250 dollars. What was the subtotal, in dollars?

A) 200.00

B) 217.39

C) 197.63

D) 187.50

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38. **PS-180** · Hard · Multiple choice · **No calculator needed**

A club has 40 members, and 30% of the members are girls. How many girls must join the club, with no one leaving, so that girls make up 60% of the members?

A) 20

B) 12

C) 30

D) 28

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39. **PS-181** · Hard · Multiple choice · **Calculator**

A worker spends 30% of her monthly salary on rent, then 25% of the remaining amount on food, then 40% of what is left after that on savings. She has $3{,}780$ dollars left. What is her monthly salary, in dollars?

A) $75{,}600$

B) $7{,}200$

C) $12{,}000$

D) $6{,}300$

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40. **PS-182** · Hard · Multiple choice · **Calculator**

Two positive numbers a and b are in the ratio $a : b = 9 : 5$. The number b is what percent less than a, to the nearest tenth of a percent?

A) 44.4%

B) 80.0%

C) 55.6%

D) 28.6%

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41. **PS-183** · Hard · Multiple choice · **No calculator needed**

The number y is $p\%$ less than the positive number x, where $0 < p < 100$. Which expression is equal to x?

A) $\dfrac{(100 - p)y}{100}$

B) $\dfrac{100y}{100 + p}$

C) $y\left(1 + \dfrac{p}{100}\right)$

D) $\dfrac{100y}{100 - p}$

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42. **PS-184** · Hard · Multiple choice · **No calculator needed**

Fresh dates are 80% water by mass. After drying, the dates are 50% water by mass. How many kilograms of dried dates are produced from 60 kilograms of fresh dates?

A) 30

B) 24

C) 42

D) 12

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43. **PS-185** · Hard · Multiple choice · **No calculator needed**

In a survey, exactly 37.5% of the people answered "yes," exactly 20% answered "no," and the rest were undecided. What is the least possible number of people surveyed?

A) 5

B) 40

C) 8

D) $1{,}000$

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44. **PS-186** · Hard · Multiple choice · **No calculator needed**

At a school, 85% of the boys and 60% of the girls passed an exam, and 70% of all the students passed. What percent of the students at the school are boys?

A) 25%

B) 50%

C) 60%

D) 40%

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45. **PS-187** · Medium · Multiple choice · **Calculator**

A runner's pace is 6 minutes 15 seconds per kilometer. What is the runner's speed, in kilometers per hour?

A) 9.6

B) 6.25

C) 9.76

D) 0.16

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46. **PS-188** · Hard · Multiple choice · **No calculator needed**

A pipe can fill an empty tank in 6 hours, and a drain can empty the full tank in 9 hours. Starting with the tank empty, the pipe and the drain are both open for 4 hours, and then the drain is closed. How many more hours does it take the pipe to fill the tank?

A) 2

B) $\frac{26}{3}$

C) $\frac{14}{3}$

D) 14

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47. **PS-189** · Hard · Multiple choice · **No calculator needed**

A crew of 12 workers can finish a job in 20 days. After the crew has worked for 5 days, 3 more workers join, all working at the same rate. How many days in total does the job take?

A) 16

B) 25

C) 17

D) 12

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48. **PS-190** · Hard · Multiple choice · **No calculator needed**

A driver travels from one city to another at an average speed of 60 kilometers per hour. At what average speed, in kilometers per hour, must the driver return along the same road so that the average speed for the whole round trip is 80 kilometers per hour?

A) 120

B) 106.7

C) 160

D) 100

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49. **PS-191** · Hard · Multiple choice · **No calculator needed**

A printer prints p pages every m seconds. Which expression gives the number of minutes the printer takes to print n pages?

A) $\frac{60m}{pn}$

B) $\frac{60mn}{p}$

C) $\frac{pn}{60m}$

D) $\frac{mn}{60p}$

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50. **PS-192** · Hard · Multiple choice · **Calculator**

A bank sells US dollars at 3.77 riyals per dollar. It first deducts a 1% commission from the riyals a customer pays. How many US dollars does a customer receive for $5{,}655$ riyals?

A) $1{,}515$

B) $1{,}485$

C) $21{,}319$

D) $1{,}500$

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51. **PS-193** · Hard · Multiple choice · **No calculator needed**

The data set $3, 5, 6, 8, 9, 11, 12, 46$ has one outlier, 46, which is removed. Which statement about the effect of removing 46 is true?

A) Only the mean changes.

B) Both the mean and the median decrease, and the mean decreases more.

C) The mean decreases and the median increases.

D) The mean decreases and the median stays the same.

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52. **PS-194** · Hard · Multiple choice · **No calculator needed**

A data set consists of $12, 15, 18, 21, 24, 27, 30, 33, 36, 40$. The value 40 is mistakenly recorded as 400. How do the median, the interquartile range (IQR) and the standard deviation of the recorded data compare with those of the correct data?

A) Only the median is unchanged.

B) The median, IQR and standard deviation are all unchanged.

C) The median is unchanged, but the IQR and standard deviation are greater.

D) The median and IQR are unchanged, but the standard deviation is greater.

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53. **PS-195** · Hard · Multiple choice · **No calculator needed**

Section P has 60 students and section Q has 24 students. Test-score summaries: Section P: minimum 55, first quartile 68, median 76, third quartile 84, maximum 97. Section Q: minimum 62, first quartile 77, median 83, third quartile 88, maximum 95. Which statement must be true?

A) At least 75% of the students in Q scored above the median of P.

B) The IQR of Q is greater than the IQR of P.

C) More students in Q than in P scored above 84.

D) The range of Q is greater than the range of P.

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54. **PS-196** · Hard · Multiple choice · **No calculator needed**

At a company, the median monthly salary is $9{,}000$ dollars and the mean monthly salary is $14{,}500$ dollars. Which statement must be true?

A) The most common salary is $9{,}000$ dollars.

B) The standard deviation of the salaries is $5{,}500$ dollars.

C) No more than half of the employees earn more than $14{,}500$ dollars.

D) More than half of the employees earn more than $14{,}500$ dollars.

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55. **PS-197** · Hard · SPR · **No calculator needed**

In a data set, the values 1, 2, 3, 4 and 5 occur with relative frequencies 10%, 15%, 20%, 30% and 25%, respectively. What is the median of the data set minus the mean of the data set?

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56. **PS-198** · Hard · Multiple choice · **No calculator needed**

A data set has six values. Their deviations from 40 (value minus 40) are -7, 3, 12, -5, x and 2. The mean of the data set is 42. What is the value whose deviation from 40 is x?

A) 35

B) 7

C) 47

D) 52

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57. **PS-199** · Hard · Multiple choice · **No calculator needed**

A data set of 30 values has a mean of 50 and a standard deviation of 10. The values 38 and 62 are added to the data set. Which statement is true?

A) The mean is unchanged and the standard deviation increases.

B) The mean is unchanged and the standard deviation decreases.

C) The mean and the standard deviation are both unchanged.

D) The mean and the standard deviation both increase.

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58. **PS-200** · Medium · SPR · **No calculator needed**

In a data set, a value is called a low outlier if it is less than $Q_1 - 1.5 \cdot \text{IQR}$, where Q_1 is the first quartile and IQR is the interquartile range. For a data set, $Q_1 = 28$ and the third quartile is 40. What is the greatest integer value that would be a low outlier?

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59. **PS-201** · Hard · Multiple choice · **No calculator needed**

The recorded values of a data set are $18, 22, 27, 30, 33, 36, 41, 47, 52$. The value 52 should have been 25. After the correction, how do the mean and the median change?

A) The mean decreases by 3 and the median is unchanged.

B) The median decreases and the mean is unchanged.

C) The mean decreases by 27 and the median decreases by 3.

D) The mean and the median both decrease by 3.

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60. **PS-202** · Hard · Multiple choice · **No calculator needed**

A company randomly selected 200 of its employees and gave them new office chairs. Their average productivity rose 8% during the next 3 months. Which conclusion is best supported?

A) The new chairs caused the rise for all employees.

B) The rise cannot be attributed to the chairs, because there was no comparison group.

C) The new chairs had no effect on productivity.

D) The new chairs caused the rise for the 200 employees only.

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61. **PS-203** · Hard · Multiple choice · **No calculator needed**

In an experiment, garden plots were randomly assigned to receive 0, 20 or 40 grams of a fertilizer. The plots given 40 grams had the highest mean yield. Which conclusion is best supported?

A) Doubling the fertilizer doubles the yield.

B) For plots like these, increasing the fertilizer from 0 to 40 grams caused higher yields.

C) No causal conclusion is possible.

D) Using 80 grams would give an even higher yield.

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62. **PS-204** · Hard · Multiple choice · **No calculator needed**

A survey was mailed to $1{,}200$ randomly selected households, and 180 surveys were returned. Of these, 72% supported a new fee. What is the main reason the 72% may not represent all households?

A) There was no random assignment to treatments.

B) The households were selected at random.

C) The sample of $1{,}200$ is too small.

D) Many selected households did not respond, and those who responded may differ from those who did not.

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63. **PS-205** · Medium · Multiple choice · **No calculator needed**

An observational study found that children from homes with more books tend to have higher reading scores. Which of the following is the most plausible confounding variable?

A) Parents' education level

B) The scale used for the reading test

C) The number of pages per book

D) The number of children in the study

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64. **PS-206** · Medium · Multiple choice · **No calculator needed**

To estimate the average weekly exercise time of adults in a city, a researcher randomly selected 400 people from the membership lists of the city's gyms. Why is the estimate likely to be biased?

A) The sample size is too small.

B) The selection was random, so the estimate is not biased.

C) The list leaves out adults who are not gym members.

D) There was no random assignment.

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65. **PS-207** · Medium · Multiple choice · **No calculator needed**

A teacher randomly selected 50 students and asked each one, face to face, "Have you ever copied homework?" Of the students, 6% answered yes. Which is the most likely problem with this result?

A) The sample was not selected at random.

B) There was no control group.

C) Students may give untrue answers because the question is sensitive and asked in person.

D) A sample of 50 students is too large.

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66. **PS-208** · Medium · Multiple choice · **No calculator needed**

For a data set, the correlation coefficient between height in centimeters and mass in kilograms is $r = 0.72$. The heights are converted to inches, the masses to pounds, and the two variables are swapped between the axes. What is the correlation coefficient for the converted data?

A) 0.72

B) -0.72

C) $0.72 \cdot \dfrac{2.2}{2.54}$

D) $\dfrac{1}{0.72}$

67. **PS-209** · Hard · Multiple choice · **No calculator needed**

For data set P, the line of best fit is $\hat{y} = 4.2x + 3$ and the correlation coefficient is 0.41. For data set Q, the line of best fit is $\hat{y} = 0.6x + 10$ and the correlation coefficient is 0.93. Which statement is true?

A) The associations cannot be compared because the slopes are different.

B) P has the stronger linear association because its slope is greater.

C) Q has a weaker association because its slope is less than 1.

D) Q has the stronger linear association.

68. **PS-210** · Hard · Multiple choice · **No calculator needed**

A model for a data set is $\hat{y} = 2.5x + 10$. The data points are $(2, 17)$, $(4, 19.5)$, $(6, 24)$, $(8, 31)$ and $(10, 34.5)$. For how many of the data points does the model underestimate the actual value of y?

A) 5

B) 2

C) 3

D) 1